

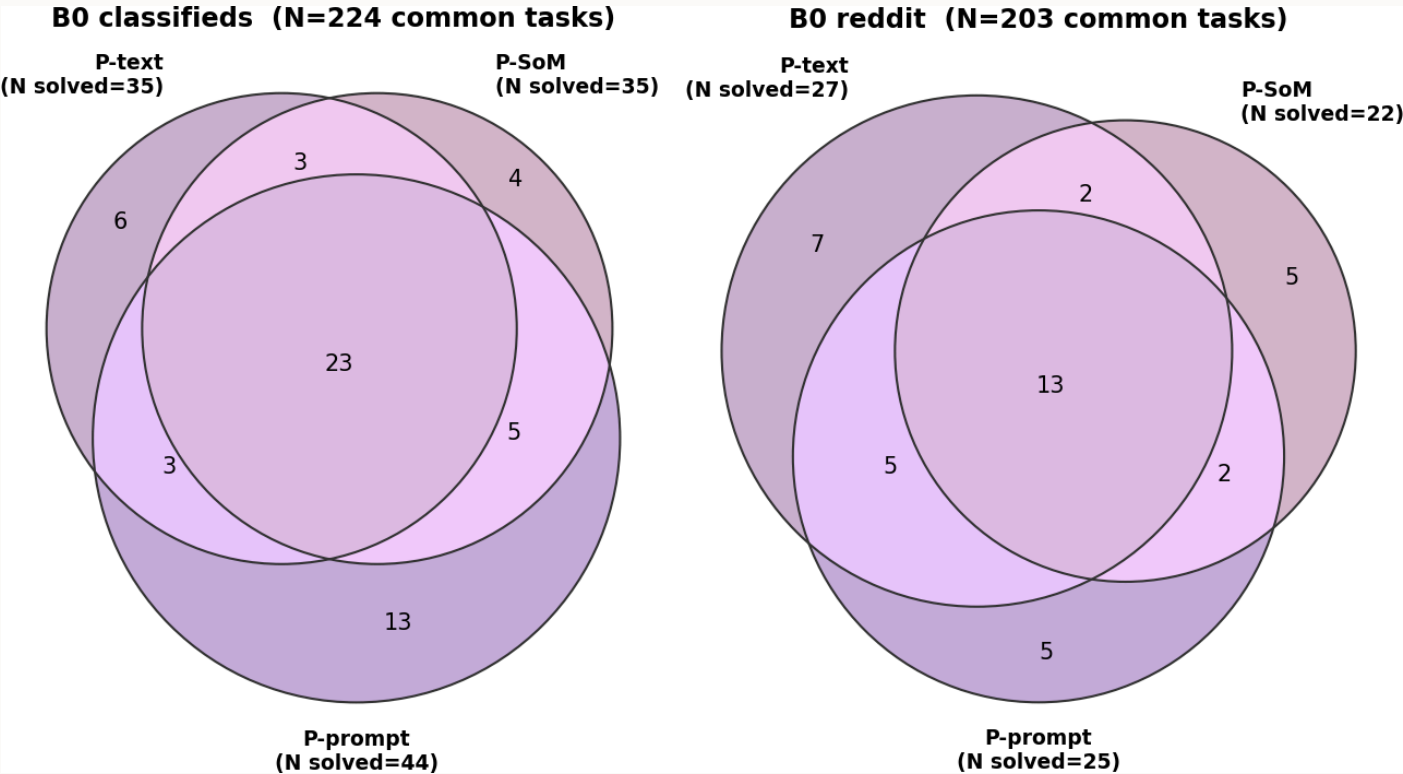
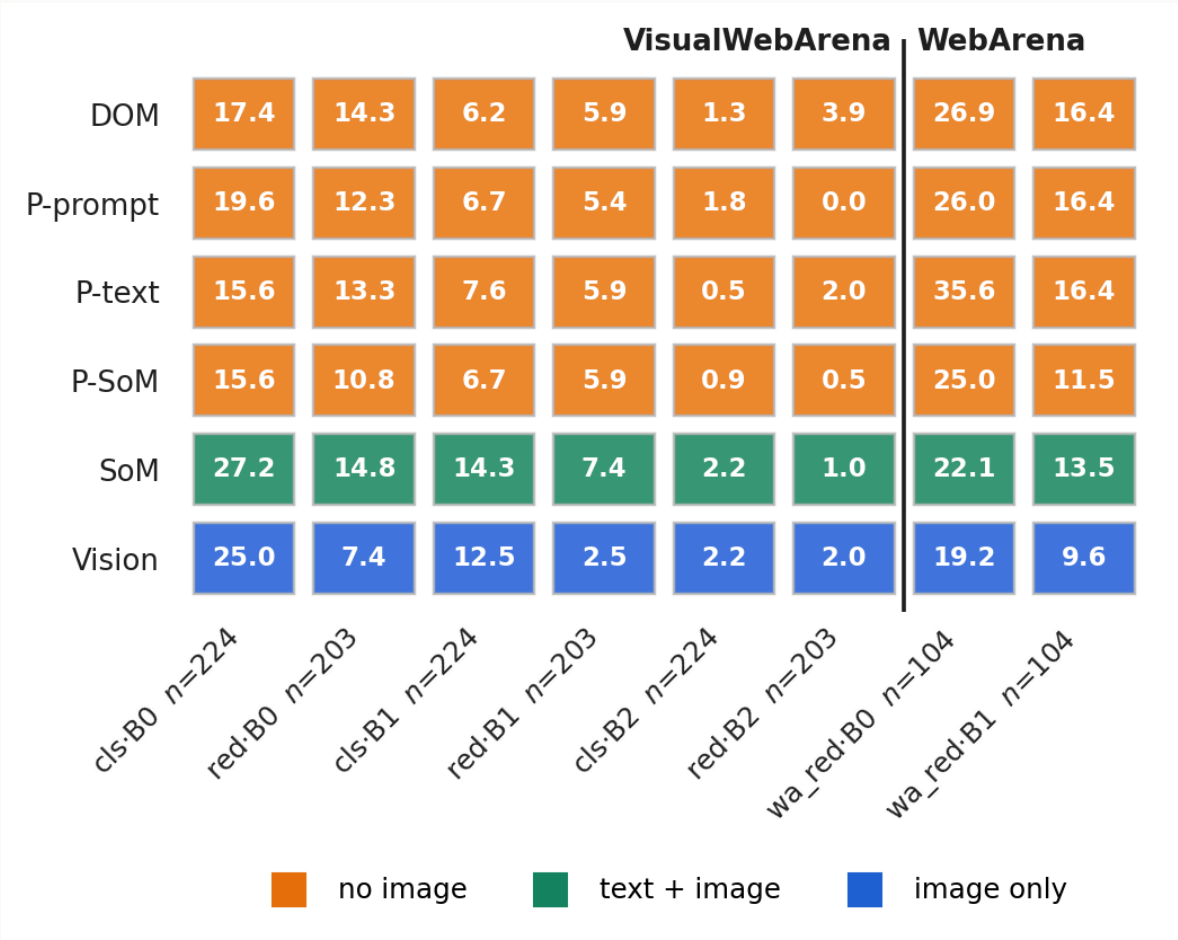
MSC THESIS · REALM WORKSHOP, EMNLP 2026

When is expensive perception worth paying for?

ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

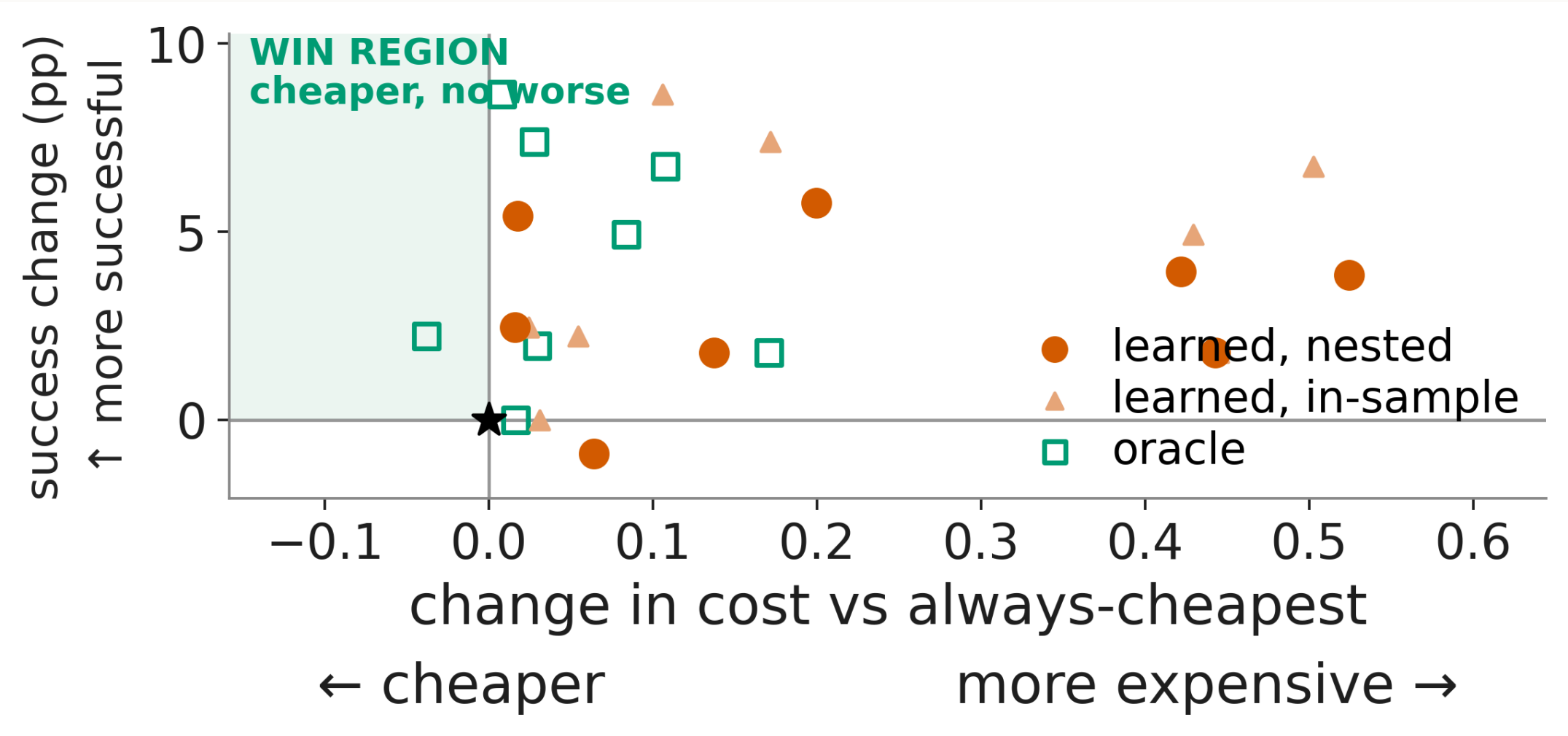
Same task, three ways of seeing it, three different bills.

Hindsight says choosing pays; a rerun says how much of that is noise.



Hindsight: **+3.45 to +16.35** in 100 vs the best single view · a rerun flips **10–14%**, buys **2.0–7.6**

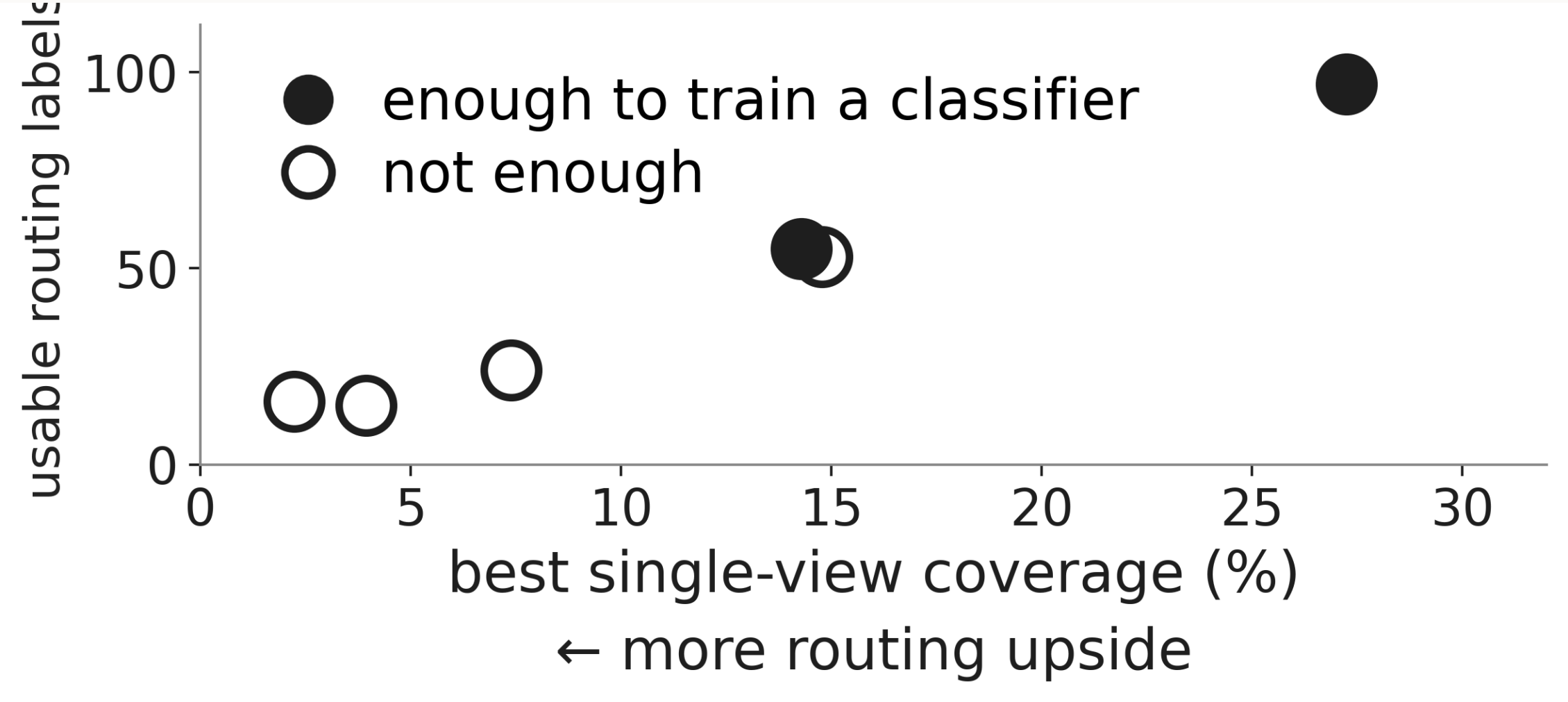
Learned routers buy success only by spending more.



Win region = cheaper and no worse. Learned: 0 of 8. Perfect hindsight: 1 of 8.

A “WHICH VIEW” EXAMPLE EXISTS ONLY WHEN THE TASK IS SOLVED

More routing upside, less usable training signal.



Base success 2–36% · 15–97 usable examples per setting · enough to train in 2 of the 6 · failing settings need at least 2.1–4.2× more tasks

NOT CLAIMED · ASKED

Improve the agent first, collect reliable examples, then learn when to look.

NOT CLAIMED

- ✗ routing is unlearnable
- ✗ always-cheapest is a floor
- ✗ live runs are scored

THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use, and why?
- ✓ lend us a stronger agent

Look, read, or both?

Hindsight says: choose.

Nothing trained knows in advance.



The board is by the window — try your own task, live in all three views.

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BACKUP — TASK 130, RECORDED STEPPING, 39 S